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Original article

Prediction of RNA editing sites and genome-wide characterization of PERK gene family in maize (*Zea mays* L.) in response to drought stress



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ABSTRACT

Objectives: Inadvertent climate changes continuously threatening the crops production and thus affecting the livelihood of peoples across the world. The maize production at world level has severely been hampered by the drought stress. Proline-rich extensin-like receptor kinases (PERKs) are considered among the sub-class of plants larger protein family, receptor kinases. Member of PERK gene family play significant role in both abiotic and biotic stress and in various plant metabolic activities and pathways.

Methods: As of now, no comprehensive research is reported for PERK genes in maize. We have performed a genome wide *in-silico* analysis and identify twenty-three PERK genes in maize. We performed phylogenetic analysis, sequence logos, motif analysis, promoter analysis, chromosomal and subcellular localization, synteny and expression analysis using RNA seq data under drought stress. We also predict RNA editing sites in mitochondrial and chloroplast genome.

Results: Phylogenetic study of PERK genes from eight different plant species divided into four distinct clades. Four subclasses group of ZmPERKs were observed based on domain organization, motif pattern, and phylogenetic analysis. The exon–intron arrangement of the ZmPERK were conserved among members of the same subclasses. In the promoter region different *cis*-elements were found those were involved in the growth and development, as well as light and stress response. Through gene duplication analysis it was observed that segmental duplications in ZmPERKs played major role in maize evolution. The Ka/Ks ratios indicated that most ZmPERK genes during the evolution have experienced strong purifying selection. The conversion of cytosine (C) to uracil (U) was observed in all predicted editing sites (U). These transitions were mostly based on changes in the first and second codon bases. The *in-silico* expression analysis of transcriptome data revealed the differential expression of ZmPERK genes in response to drought stress and oil content accumulation.

Conclusion: The current study provides base information on the PERK gene family in maize. Our findings can serve as a reference for further functional analysis of ZmPERKs. These genes can be further explored and used in breeding program to develop cultivars resilient to drought stress.

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1. Introduction

Maize is considered as important crop which is grown for food and feed around the globe. Abiotic stress especially drought stress severely affected the maize production at world level. Plant breeders are considering the drought stress as one of the most important abiotic stress that causing the hindrance in getting the higher grain yield in different crops especially in maize (Liu and Qin, 2021).

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Hence it is dire need of the time that breeders should tailor their modern varieties with novel traits that have capacity to buffer the drastic effects of the abiotic stresses especially drought stress. The advent of modern genotyping techniques especially the next generation sequencing (NGS) has revolutionized the field of genetics, furthermore the release of genome dataset has made a substantial increment in developing the strategies to quip the plants with novel traits (Leng and Zhao, 2020). In plants the receptor-like kinases (RLKs) having similar structure are considered as large superfamily of proteins. In this group the PERKs (Proline rich extension like receptor kinases) gene family is also included. Plant Species such as Arabidopsis and rice comprises large gene family of receptor kinase. There is 600 members are reported in arabidopsis for the receptor kinase family and their analogous have been found in around 20 different species (Morris and Walker, 2003). These kinase family play have been found playing a crucial role in the growth and development phase of plants and also defensive mechanisms (Shiu and Bleeker, 2001, Morris and Walker, 2003, Shiu et al., 2004). Role of most of the members of receptor kinase is still unknown. Receptors like kinase are the protein comprising extracellular, carboxyl terminal and intercellular domain with putative amino terminal (see Fig. 1).

Depending on their extracellular domain, receptor kinases interact to a wide range of substances for example, carbohydrates and cell wall components. This domain organization have much resemblance to the animal receptor tyrosine kinases (Shiu, 2001). Receptor kinases have specific and extracellular domains, for example, leucine-rich repeat (LRR) and proline-rich extension-like receptor kinases. Gene duplication and functional redundancy also reported among these different classes of receptor kinase (Champion et al., 2004). The CLV1 and ERECTA receptor kinase is the evidence of existence of functional redundancy (Diévar et al., 2003, Shpak et al., 2003, Shpak et al., 2004). PERK gene family of Arabidopsis have maximum sequence identity to Brassica napus and such as PERK1 of Arabidopsis is much like PERK1 of Brassica napus. Researchers have reported fifteen PERK genes in the Arabidopsis yet their functions still need to be characterize (Silva and Goring, 2002, Nakhamchik et al., 2004, Bai et al., 2009). In Arabidopsis the PERK1 is identified which do functions in response to any wound that occur in the plasma membrane (Silva and Goring, 2002). Likewise, PERK4 is predicted as key regulator for Ca²⁺ sig-

naling that contributes in production of abscisic acid in root (Bai et al., 2009). It is well documented that under abiotic stresses plant accumulate more calcium contents in cells to boost the production of antioxidant enzyme activity, regulate lipid peroxidation of cell membranes and stomatal apertures to mitigate the impact of stresses on plant growth (Mansfield et al., 1990; Abadi and Sepehri, 2016). The production of reactive oxygen species (ROS) decline in the presence of PERKs whereas increasing level of ROS work as a signal for root hair development (Xing et al., 2013). In an organism the first line of defense against superoxide radicals is the production of superoxide dismutase (SODs) which catalyze the superoxide radical to hydrogen peroxide and molecular oxygen. The copper/zinc SOD (Cu/Zn SOD) is catalyzed through the MAPK cascade under high light-induction. The homologous proteins like MPK3 and MPK6 in plants are detected using the anti-PERK antibodies from animals (Samuel and Ellis, 2002; Hwang et al., 2016). Environmental stresses like heat, drought, nutrients, heavy metals, pathogens, keep threatening the plants to express its fully genetic potential. It is becoming more important to scientist to reveal how plants response to internal and external stimuli. Plant sense the environmental changes through the use of cell surface receptors and initiate different signaling pathways to trigger the adaptive responses (Zhu, 2016).

Erratic climate change has become a major constraint in achieving the higher crop yield. At crop level it affects plant morphological, anatomical, and physiological attributes which ultimately results in drastic economic yield loss. Maize is a major food and feed crop grown all over the world. It rated as the world's third most significant staple grain crop (Tiware and Yadav, 2019). Characterization of PERK family in maize can help us to understand the plant molecular mechanism of tolerance against biotic and abiotic stresses. Only a few PERK genes have been characterized, and the functions of most of them is still unknown. High-throughput genome sequencing of the maize provided an excellent opportunity for genome wide analysis of genes families. In our study, we performed *in-silico* genome wide analysis of PERK genes in maize. We analyzed phylogenetic relationship between 8 species and only maize separately. Furthermore, gene structure Intron/exons, motif distribution, conserved domain analysis, sequence logos, physiochemical properties. The structural and functional importance of genes were also assessed using Ka/Ks values and synteny analysis.

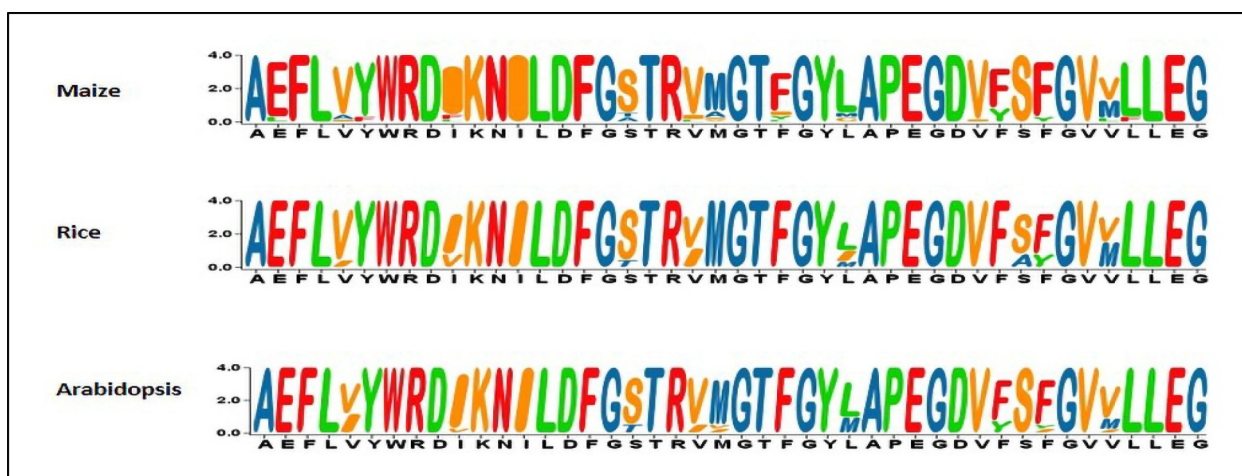


Fig. 1. Sequence logos of PERK gene family between Maize, Rice and Arabidopsis.

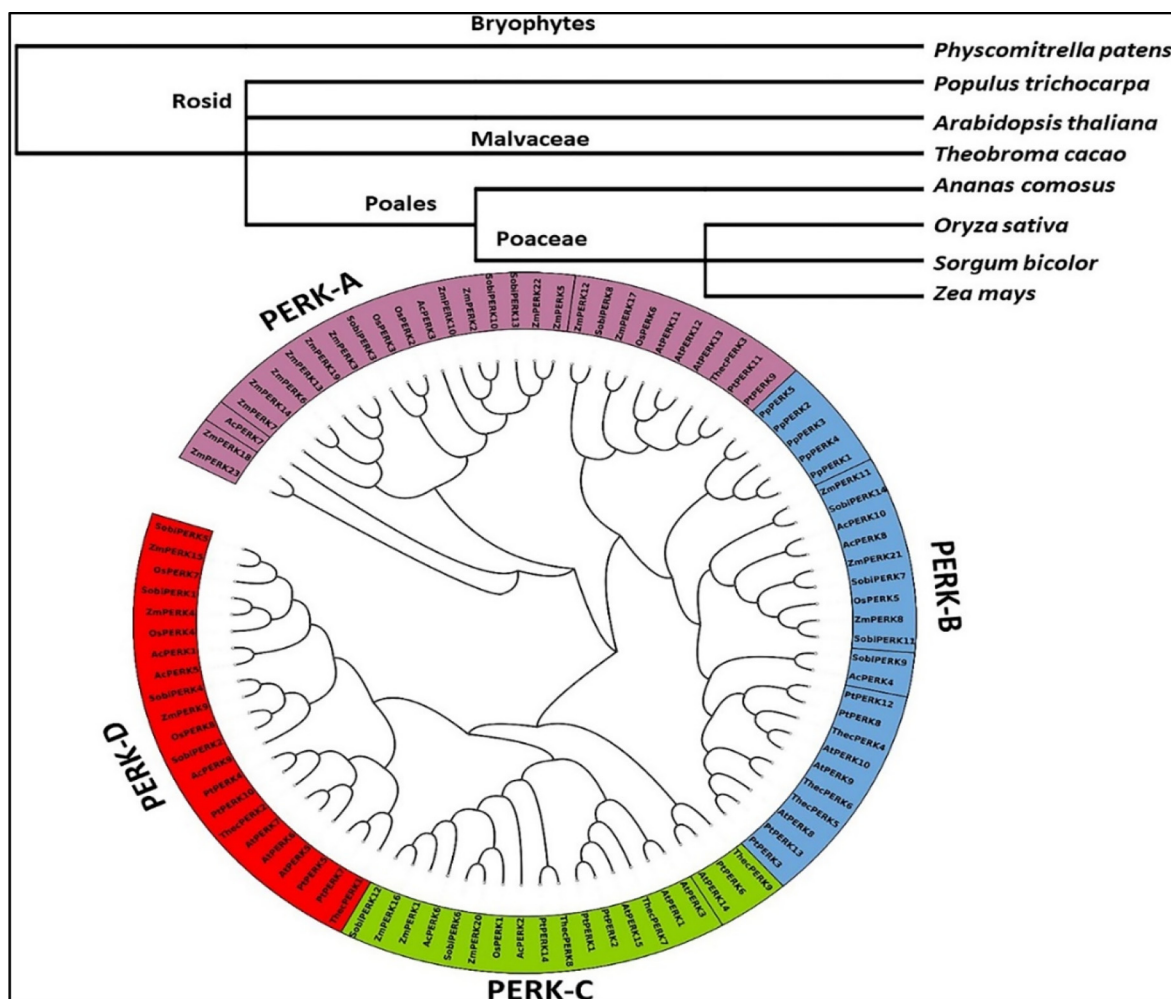


Fig. 2. PERK gene family phylogenetic tree. The major cluster of orthologous genes is distinguished with various colours (PERK-A-D).

In-silico expression analysis were also performed for these genes to predict their role and function. The present study and their result enabled us to conclude that PERK gene family play vital role in maize development and stress response (Fig. 2).

2. Materials and methods

2.1. PERK gene family identification and characterization in maize genome

Whole maize genome sequence, as well as the general feature format file (GFF3), was downloaded from the Maize Genetic and Genomic Database (Maize GDB) (<https://gamma.maizegdb.org>). For the purposes of finding the probable candidates of PERK family in maize, the online Pfam database (<https://www.sanger.ac.uk/Software/Pfam/>) was used to download the PERK domain HMM profile and then subjected as a query into Blastp (Finn et al., 2014). For all of the retrieved protein sequences, the SMART tools (<https://smart.embl-heidelberg.de/>) were used to verify the presence of the PERK domain (Letunic et al., 2015). Maize PERK gene family sequence were downloaded from maize genome database. TAIR 10 (<http://www.Arabidopsis.org>) was used to retrieve the arabidopsis sequences while all other sequences of studied organisms were retrieved from online plant database Phytozome version 11 (<https://phytozome.jgi.doe.gov/pz/portal.html>). ExPASyProtParam, (<https://us.expasy.org/tools/protparam.html>) an online web tools, were used to retrieve the physiochemical properties.

2.2. Sequence logos and phylogenetic/evolutionary analysis

The MEGA 7 software was used to find out the conserved sequences for amino acids. Sequence are aligned using ClustalW and the structure was constructed using TBtool (<https://github.com/CJ-Chen/TBtools>). Furthermore, using this software the Neighbor-Joining method was used to get the phylogenetic tree to deduce the evolutionary history (Chothia et al., 2003). The distances of the number of amino acid sites in units were measured using the poisson correction parameters (Yang et al., 2008). The Bootstrap algorithm employed with 1000 repetitions to estimate the stability of the nodes in the phylogenetic tree. Total 98 amino acid sequences were used for this analysis.

2.3. Predicted protein motifs, structure of exon/intron and conserved domain analysis

To find preserved motif of PERK protein online web server Multiple Em for Motif Elicitation (MEME) is used (<https://meme-suite.org/tools/meme>). The TBtool was used to construct the motif structure using the MEME.xml file which is obtained through MEME suite. The default parameters were as follows: motif recurrence was set to 1 per sequence; frequency of motifs was set to 10; motif width was set to 5–50 residues; and the minimum number of motif sites was set to 5. Arrangement of Exon and Intron of PERK genes was investigated by using gff3 file downloaded from maize GDB. Structure is constructed using TBtool software. Afterward, the NCBI

CDD tool (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) was used to perform the conserved domain analysis.

2.4. Chromosomal localization, gene duplication events and synteny analysis

The start and end position for each identified *ZmPERK* gene were procure from the Maize Genetic and Genomic Database (Maize GDB) and validated against the GFF3 file. Total chromosomal length is retrieved through using FASTA stat in TBtool. Finally, using MapChart v2.32 (Voorrips, 2002), the *ZmPERK* genes were spatially mapped onto the maize chromosomes. The phylogenetic tree was used to identify the putative paralogous PERK gene pairs. The resulting pairs were subjected to TBtool software to determine synonymous and non-synonymous substitution rates (Ka). To determine the nature of codon selection the Ka/Ks ratio was also calculated that allegedly occurred during evolution. Further, using the formula $T = Ks/2$ and assuming a clock rate of 6.05×10^{-9} substitutions/synonymous site/year for maize, the approximate period of duplication event was calculated (Kong et al., 2013). The genome sequence files, and gene annotation files (GFF3) of sorghum, rice and maize were used for the collinearity analysis. Required files generated using one step MC scan. TBtool software was used to visualize the results, and the parameter filtering genes in the collinearity block was set to 40.

2.5. RNA editing sites prediction, subcellular localization, and promoter analysis

RNA editing is a method in which certain cytidines in mitochondrial and chloroplast transcripts of plants are converted to uridines. The online web server like PREP-Cp (for chloroplast genes) and PREP-Mt (for mitochondrial genes) software (<https://prep.unl.edu/>) with the cutoff value to 0.8 were used in predicting the RNA editing sites (Mower, 2009). Location of genes at cellular level were also predicted using online web server softberry (<https://www.softberry.com>). In order to perform promoter analysis, the 5' upstream region of each gene of the *ZmPERK* was downloaded from NCBI (<https://www.ncbi.nlm.nih.gov/>) and the resulting file was submitted to the online database plantCARE (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) for a cis-element scan.

2.6. In-silico expression analysis

The NCBI Geodataset (<https://www.ncbi.nlm.nih.gov/gds>) was used to obtain transcriptome data (GSE136087, GSE40070). RNA-sequencing analysis was performed on seed embryos from 15 days after pollination (DAP) to 40 days after pollination (DAP) (Sekhon et al., 2013). We also retrieved drought stress expression data. Inbred line B73 maize (*Zea mays*) plants grown in the green house under well-watered and drought stress circumstances until they reached the reproductive stage (at the onset of silk emergence). Plants were hand pollinated for drought stress two to three days after irrigation was stopped, and measurements and samples were collected 24 h later for transcriptome analysis. This gene expression data was used to construct heat map.

3. Results

3.1. Identification and characterization of PERK gene family in maize genome

A systematic approach was followed to identify the *PERK* protein encoding genes. Taking the advantage of publically available

Table 1
Physiochemical properties including Gene IDs, Gene name, Chromosome No, Gene start–end position, Strand, CDS and amino acid sequence length, Protein MW, PI, GRAVY and cellular localization of *ZmPERK* genes.

Transcripts ID	Gene name	chromosome	Position Start End	Strand	CDS (bp)	Protein length AA	Protein Weight KDa	Molecular Weight KDa	PI	Grand average of hydropathicity (GRAVY)	Subcellular localization
Zm00001d037811	ZmPERK1	6	138,545,311 201,473,204	-	1299 2208	432 735	59,97726 76,90377	6.47 8.51	-0.483 -0.41	-0.483	Cell membrane/plasma membrane
Zm00001d043480	ZmPERK2	3	201,473,204 164,313,713	-	2208 1479	735 792	76,90377 53,98962	8.51 9.3	-0.41 -0.633	-0.41	Cell membrane/plasma membrane
Zm00001d011908	ZmPERK3	8	164,313,713 288,176,999	-	1479 1761	792 588	53,98962 61,80629	9.3 9.13	-0.633 -0.475	-0.633	Cell membrane/plasma membrane
Zm00001d034257	ZmPERK4	1	288,176,999 113,763,279	-	1761 1488	588 495	61,80629 53,51192	9.13 6.41	-0.475 0.37	-0.475	Cell membrane/plasma membrane
Zm00001d030218	ZmPERK5	1	113,763,279 122,414,822	+	1488 1524	495 507	53,51192 56,60236	6.41 9.12	0.37 -0.039	0.37	Cell membrane/plasma membrane
Zm00001d041476	ZmPERK6	3	122,414,822 95,798,411	-	1524 1797	507 598	56,60236 63,45431	9.12 8.31	-0.039 -0.228	-0.039	Cell membrane/plasma membrane
Zm00001d020148	ZmPERK7	7	95,798,411 47,428,910	-	1797 1788	598 595	63,45431 63,40166	8.31 8.91	-0.228 -0.311	-0.228	Cell membrane/plasma membrane
Zm00001d035774	ZmPERK8	6	47,428,910 31,103,531	-	1788 1752	595 583	63,40166 61,77299	8.91 6.5	-0.311 -0.461	-0.311	Cell membrane/plasma membrane
Zm00001d028337	ZmPERK9	1	31,103,531 179,562,397	-	1752 1233	583 410	61,77299 45,3461	6.5 9.17	-0.461 -0.463	-0.461	Cell membrane/plasma membrane
Zm00001d012743	ZmPERK10	8	179,562,397 126,133,881	+	1233 1671	410 556	45,3461 58,84482	9.17 5.68	-0.463 -0.401	-0.463	Cell membrane/plasma membrane
Zm00001d037464	ZmPERK11	6	126,133,881 150,848,228	+	1671 2052	556 683	58,84482 71,52957	5.68 8.67	-0.401 -0.488	-0.401	Cell membrane/plasma membrane
Zm00001d011450	ZmPERK12	8	150,848,228 149,652,422	-	2052 2877	683 958	71,52957 100,93417	8.67 5.24	-0.488 -0.048	-0.488	Cell membrane/plasma membrane
Zm00001d026668	ZmPERK13	10	149,652,422 28,932,486	-	2877 1002	958 333	100,93417 36,4539	5.24 8.97	-0.048 -0.07	-0.048	Cell membrane/plasma membrane
Zm00001d049391	ZmPERK14	4	28,932,486 240,835,869	+	1002 1614	333 535	36,4539 56,9581	8.97 9.15	-0.07 -0.383	-0.07	Cell membrane/plasma membrane
Zm00001d007848	ZmPERK15	2	240,835,869 114,332,739	-	1614 1989	535 662	56,9581 69,62817	9.15 8.72	-0.383 -0.523	-0.383	Cell membrane/plasma membrane
Zm00001d010421	ZmPERK16	8	114,332,739 111,296,289	+	1989 1789	662 662	69,62817 75,50927	8.72 6.3	-0.523 -0.515	-0.523	Cell membrane/plasma membrane
Zm00001d037066	ZmPERK17	6	111,296,289 191,554,665	+	1789 1401	662 466	75,50927 51,97133	6.3 6.2	-0.515 0.159	-0.515	Cytoplasm
Zm00001d031482	ZmPERK18	1	191,554,665 155,343,261	+	1401 1476	466 491	51,97133 53,59911	6.2 9.15	0.159 -0.608	0.159	Cell membrane/plasma membrane
Zm00001d042185	ZmPERK19	3	155,343,261 1,451,510	-	1476 1125	491 374	53,59911 57,67767	9.15 8.41	-0.608 -0.488	-0.608	Cell membrane/plasma membrane
Zm00001d039311	ZmPERK20	3	1,451,510 28,623,779	-	1125 2076	374 691	57,67767 72,32502	8.41 8.44	-0.488 -0.278	-0.488	Cell membrane/plasma membrane
Zm00001d040127	ZmPERK21	3	28,623,779 163,090,768	-	2076 2691	691 896	72,32502 93,90373	8.44 8.61	-0.278 -0.3	-0.278	Cell membrane/plasma membrane
Zm00001d038708	ZmPERK22	6	163,090,768 171,828,226	-	2691 1401	896 466	93,90373 51,97133	8.61 6.2	-0.3 0.159	-0.3	Cytoplasm
Zm00001d039176	ZmPERK23	6	171,828,226	+	1401	466	51,97133	6.2	0.159	0.159	Cytoplasm

genome of maize, after removing redundant genes we excavated 23 *PERK* genes in maize genome and named them as *ZmPERK1* to *ZmPERK23*. The *PERK* gene family comprises 15 *Arabidopsis thaliana* genes, 8 *Oryza sativa* genes, 14 *Populus trichocarpa* genes, 14 *Sorghum bicolor* genes, 9 *Theobroma cacao* genes, 23 *Z. mays* genes, 10 *Ananas comosus* genes, and 5 *Physcomitrella patens* genes. The biophysical properties of the *ZmPERK* family genes were then determined, including genes ID, start and end positions of genes on chromosomes, polarity of strand, length of CDS sequence (bp), length of protein sequence (aa), protein molecular weights (MW), isoelectric points (pI), and predicted subcellular localization of *ZmPERK* genes. Table 1 presents all additional estimated biophysical properties. *ZmPERK* proteins had a peptide length ranged from 432 to 958 amino acids, with an average of 695 A.A (Table 1). The PI (Isoelectric point) of maize *PERKs* varied between 6.2 and 9.134, while the molecular weight ranged between 36.45 and 100.93 kDa, with an average of 68.69 kDa. The length of nucleotide, amino acid sequences varied greatly, indicating that the *ZmPERK* genes are highly complex, implying a high level of complexity.

3.2. Sequence logos and phylogenetic/evolutionary analysis

Sequence logos analysis provide more comprehensive information for sequence similarities, significant alignment aspects, and sequence conservation patterns. To check *PERK* family evolution, we generate sequence logos and results showed that this family remained conserved throughout evolution. For comparison study, the protein sequences of maize, rice and *Arabidopsis* were used. The results reveal that consensus sequence residues were highly conserved, and there was no compositional bias seen across any specie. These results help in discover and analyze and evaluate *PERK* gene family protein sequence across the species.

Phylogenetic tree serves an important way to understand the evolutionary relationships pathways. In our study we created phylogenetic tree of *PERK* genes to depict the evolutionary relationships. The phylogenetic or evolutionary analysis revealed the oldest plant lineage, of the *PERK* gene family as its members were found in *Ananas comosus* (angiosperm), *Physcomitrella patens* (bryophytes), dicots (*Arabidopsis thaliana*, *Theobroma cacao*, *Populus trichocarpa*), and monocots (*Oryza sativa*, *Sorghum bicolor* and *Z. mays*). These findings suggested that these genes evolved in ancient land plants, and that probable orthologous genes can be found across the plant kingdom. The *PERK* genes were characterized by 29 members in the *PERK-A* clade, 26 members in *PERK-B*, 19 members in *PERK-C*, and 22 members in *PERK-D* in the phylogenetic study. *PERK* genes were randomly distributed in all four clades from dicot, monocot, and bryophytes plant species, indicating that these genes evolved after the split of bryophytes. This finding showed that *PERK* genes possibly expanded and diversified after the radiation of these different species. These evolutionary linkages can facilitate the identification of orthologous genes and help to accelerate their functional characterization.

3.3. Predicted protein motifs, structure of exon/intron and conserved domain analysis

The 23 *ZmPERK* protein sequences were classified into four subfamilies using a rectangular phylogenetic tree (subfamily I, II, III, IV). 10 members were found in the Subfamily I, followed by subfamily II (5), subfamily III (4), and subfamily IV (4) (Fig. 3A). In addition, we examined the conserved motifs using MEME software to further investigate the diversity of *ZmPERK* protein family (Fig. 3). In this study, total 10 motifs were found (Table S3). All the gene exhibits same motif pattern. The type, order, and number of motifs were consistent within a subfamily, but varied across

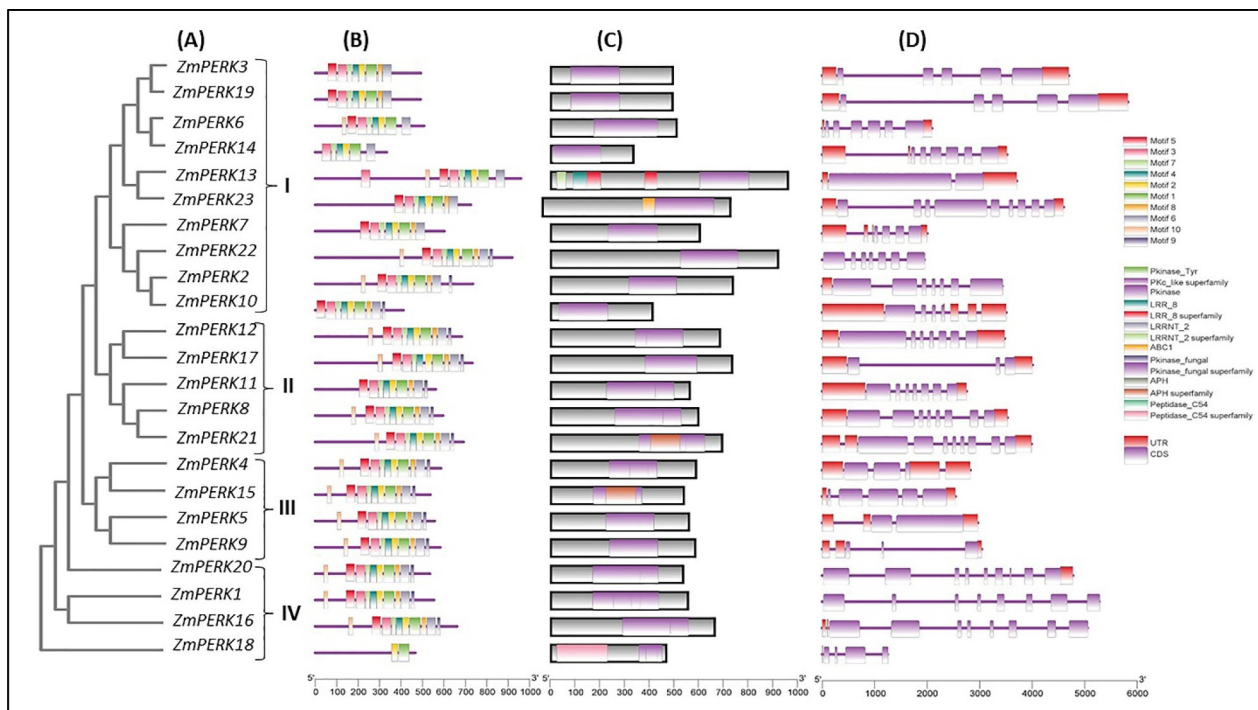


Fig. 3. A: Phylogenetic tree-based categorization of *ZmPERK* genes. An un-rooted phylogenetic tree based on full-length peptide sequences (*ZmPERK*) was generated. Classification is shown based on a phylogenetic tree using differences into groups. 3B: Motif pattern of *ZmPERK* genes 3C: Conserved domains of maize *PERK* protein 3D: Exon-intron structure analyses of *ZmPERK* genes. The purple line represents introns, while the purple boxes represent exons.

subfamilies. The patterns of ZmPERK protein motif distribution revealed that conserved distribution patterns existed for similar motifs. Domains 1, 2 and 3 represent the distinctive protein kinase-binding domain that is found in all 23 ZmPERK proteins (Fig. 3C). Similarly, Fig. 3D depicts the relative lengths of introns and exon sequence conservation within each ZmPERK gene in maize. A gene's biological function is linked to the distribution of exons and introns. All these genes contain exons ranged between 2 and 10. The findings demonstrated obvious conservation, laying the groundwork for functional conservatism and guiding future functional research.

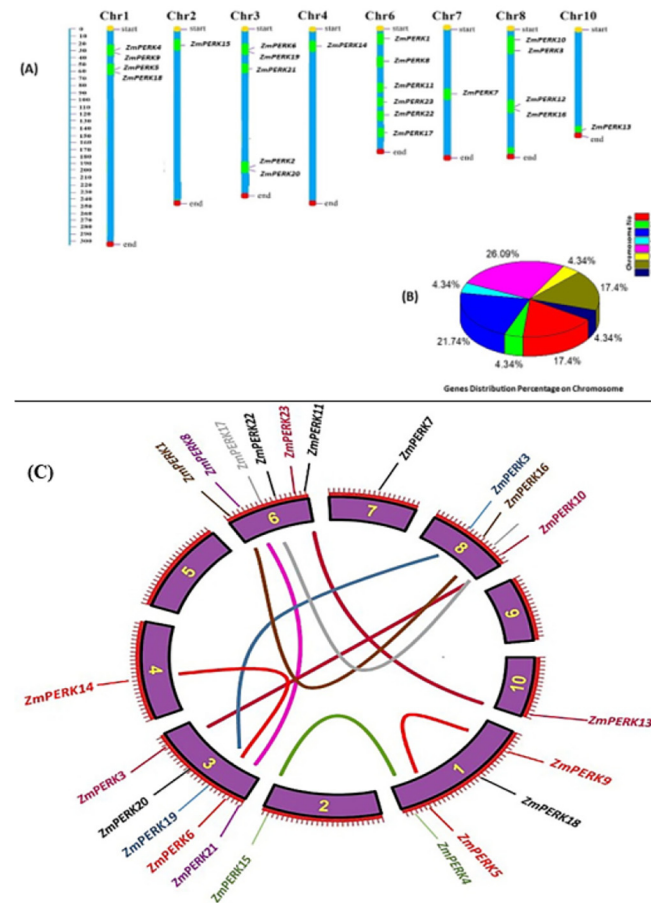


Fig. 4. A: Distribution of 23 ZmPERK genes on their respective chromosomes. B: Pie chart representing percentage of genes present on chromosome. C: Pictorial representation of paralog gene pairs on chromosome indicating the type of duplication either tandem or segmental.

Table 2
Gene duplication event along with Ka/Ks ratio and time of evolution MYA.

Gene I	Gene II	Ka	Ks	Ka/Ks	Type of Duplication	T = Ks/2λ
ZmPERK3	ZmPERK19	0.022323415	0.212693	0.104956101	Segmental	6.97
ZmPERK6	ZmPERK14	0.241933478	0.521139	0.658737178	Segmental	4.21
ZmPERK13	ZmPERK23	0.05503852	0.043242	0.120303727	Segmental	1.31
ZmPERK2	ZmPERK10	0.146633906	0.370521	0.395750292	Segmental	1.21
ZmPERK12	ZmPERK17	0.45049383	1.189433	0.378746589	Segmental	3.901
ZmPERK8	ZmPERK21	0.222577445	2.176048	0.102285191	Segmental	7.13
ZmPERK4	ZmPERK15	0.291936478	0.54189	0.538737178	Segmental	1.77
ZmPERK5	ZmPERK9	0.070311852	0.062761	0.160303727	Tandem	2.05
ZmPERK1	ZmPERK16	0.028134859	0.206977	0.135932012	Segmental	6.788

Non-synonymous and synonymous substitutions are designated by Ka and Ks, respectively.

3.4. Chromosomal localization, gene duplication events and synteny analysis

Each ZmPERK gene's genomic DNA sequence was analyzed in the maize genome database using BLASTn to establish its location, and MapChart was used to visualize the position of identified ZmPERK members on their respective chromosomes. The chromosome map revealed that 23 PERK genes were dispersed out over 8 of the 10 chromosomes. (Fig. 4A). The most ZmPERK genes were found on chr06, which had six members, followed by chr01, 03, and 08, which had 4, 5, and 4 members, respectively. While chromosomes 2, 4, 7, and 10 each had only one gene. Zmchr06 had the most PERK genes (26.08%), followed by Zmchr3 (21.73%), Zmchr1, and Zmchr8 (17.39%), while Zmchr02, Zmchr04, Zmchr06, Zmchr07, and Zmchr10 had the lowest percentage (4.34%) (Fig. 4B).

Gene duplications, either whole-genome or segmental, as well as tandem duplications, are critical for gene family evolution. Although it has been proven that segmental and tandem duplications play a key role in gene family evolution in all plants specie (Cannon et al., 2004). To investigate ZmPERK gene duplications and evolutionary processes in maize, we identified 09 pairs of probable paralogous genes using the maize PERK phylogenetic tree. It is well documented facts that tandem duplication observed when paralogous genes are present on the same chromosome, whereas segmental duplication arise when paralogous genes are located on distinct chromosomes (Panchy et al., 2016). All the paralogous gene pair appeared to have evolved by segmental duplication except one (ZmPERK5-ZmPERK9) which evolved through tandem duplication indicating that the evolution of PERK genes appears to have been dominated by segmental duplications in maize. (Fig. 4C) Segmental duplication is the primary force that drives the evolution of a gene family. The estimated time of divergence for paralogous gene pairs was determined using synonymous (Ks) and non-synonymous (Ka) substitution rates. The Ka/Ks ratios for all paralog ZmPERK varied between 0.10 and 0.65 (Table 2). It indicates that purifying selection may have been performed on codons in the development and proliferation of parallel PERK genes in maize.

Multiple collinearity scan tool was used to find orthologous genes among genomes of maize, Sorghum, and rice to further understand the Synteny links of ZmPERK genes with these plant species. (Fig. 5). 18 pairs of collinearity genes of PERK gene family between maize and rice whereas sixteen pairs in maize and sorghum were observed in the synteny analysis (Table S2). Gene IDs of all collinear genes is given supplemental file. According to these findings, the collinearity between maize and sorghum is significant as compared to the collinearity values between maize and rice furthermore, these PERK genes in maize derived from a common ancestor.

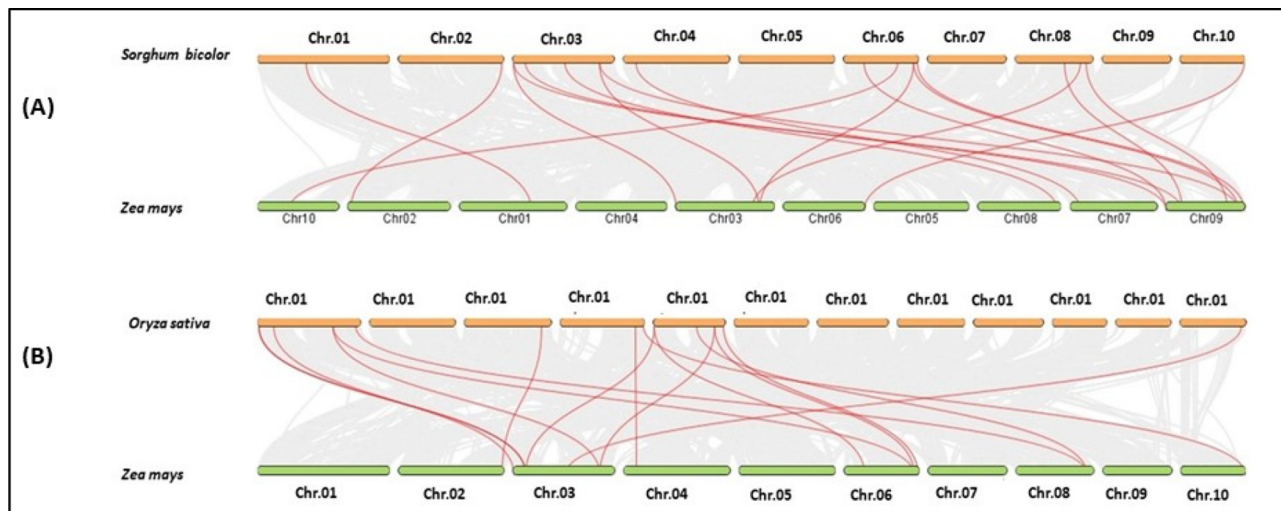


Fig. 5. Collinearity analysis of maize, rice, and sorghum. (A) Collinearity analysis of all chromosomes reveals duplicated *PERK* genes in maize and sorghum. The lines connect the pairs of duplicated genes. (B) The collinearity study of maize and rice chromosomes. The *PERK* genes are represented by the red flags on distinct chromosomes.

3.5. Prediction of RNA editing sites, subcellular localization, and promoter analysis

The Prep-CP and Prep-Mt prediction tools were used to find the RNA editing sites of *ZmPERK* chloroplast and mitochondrial genes, respectively. In chloroplast genes, 196 RNA editing sites were predicted (Table 3A) and 268 in mitochondrial genes (Table 3B). All predicted editing sites in the chloroplast and mitochondrial genomes were distributed among 23 genes, with an average of 8.5 and 11.26 editing sites per gene, respectively. The chloroplast gene *ZmPERK1* contains maximum RNA editing sites (12) while minimum editing sites (5) were predicted in *ZmPERK14*. Similarly, mitochondrial gene *ZmPERK12* contain maximum editing sites (21) while minimum sites (7) were predicted in *ZmPERK14*. The position of RNA editing sites was further explored, and it was observed that all the predicted sites were based on first and second codon base changes. At the third codon base, we couldn't find any site for RNA editing. The transition of cytosine → uracil (C-U) seems to be present in all editing sites, resulting in amino acid substitutions. Eleven type of amino acid change found in chloroplast and mitochondrial genes (Fig. 6A). Amino acid conservation caused by RNA editing including A (Alanine) → V (Valine), T (Threonine) → I (Isoleucine), H (Histidine) → Y (Tyrosine), P (Proline) → S (Serine), P (Proline) → L (Leucine), R (Arginine) → C (Cysteine), S (Serine) → F (Phenylalanine), R (Arginine) → W (Tryptophan), P (Proline) → F (Phenylalanine), S (Serine) → L (Leucine), T (Threonine) → M (Methionine), L (Leucine) F (Phenylalanine).

Location of gene at cellular level was also determined. Results indicated that 21 of the 23 *ZmPERK* proteins were localized to the plasma membrane, while two (*ZmPERK18* and *ZmPERK23*) were localized to the cytoplasm, Table 1 contains the details of these parameters. The promoter region, which is located upstream of the start codon area, controls gene transcription. Understanding gene regulation and function requires a thorough examination of *cis*-elements (Higo et al., 1999). We discovered and classified *cis*-acting factors in the upstream region of the *ZmPERK* genes. (See Table S4) The *cis*-elements were categorized based on their roles in growth and development, as well as light and stress actions. The upstream region of *ZmPERK* genes contained *cis*-acting factors

like MeJA responsive, MYB-binding sites associated with light responsive elements, ABA responsive elements, defense, stress, low temperature, gibberellin acid (GA), and salicylic acid (SA)-responsive elements, as per the promoter analysis results (Fig. 6B). The promoters of the *ZmPERK* gene have the most MeJA responsiveness elements, where they were found in 296 promoters. There were 166 light responsive elements, 88 ABA responsive elements, 27 GA responsive elements, 25 MYB light responsive elements, and 4 auxin responsive elements. The *cis*-element analysis showed that during abiotic stress and plant development phase the *ZmPERK* genes could respond.

3.6. In-silico expressions analysis

Expression patterns give information regarding the biological activities of genes because gene expression is required for optimal regulation of plant growth and development. We looked examined the expression patterns of the *ZmPERK* under drought stress and at different stages of seed embryo development from 15 days after pollination (DAP) to 40 days after pollination (DAP) in two distinct varieties (High oil content and low oil content) (Fig. 7A). Under drought stress 9 genes were upregulated in both tissues leave and cob (*ZmPERK2*, *ZmPERK 3*, *ZmPERK 8*, *ZmPERK 10*, *ZmPERK 14*, *ZmPERK 16*, *ZmPERK 19*, *ZmPERK 20*) 4 genes (*ZmPERK 1*, *ZmPERK 7*, *ZmPERK 18*, *ZmPERK 21*) only upregulated in leave tissue and 1 gene *ZmPERK 18* upregulated in cob (Fig. 7B). While Seven genes downregulated (*ZmPERK 4*, *ZmPERK 5*, *ZmPERK 11*, *ZmPERK 12*, *ZmPERK 15*, *ZmPERK 17*, *ZmPERK 22*) under drought stress. For oil content accumulation in embryo 9 genes shows upregulated expression pattern (*ZmPERK 1*, *ZmPERK 3*, *ZmPERK 6*, *ZmPERK 8*, *ZmPERK 13*, *ZmPERK 14*, *ZmPERK 16*, *ZmPERK 19* *ZmPERK*,20) while 12 genes show downregulated trend (*ZmPERK 2*, *ZmPERK 5*, *ZmPERK 6*, *ZmPERK 7*, *ZmPERK 9*, *ZmPERK 10*, *ZmPERK 11*, *ZmPERK 12*, *ZmPERK 17*, *ZmPERK 18*, *ZmPERK 22*, *ZmPERK 23*). Interestingly, some genes show similar expression pattern under both conditions. For example, *ZmPERK 3* *ZmPERK 8* *ZmPERK 14* *ZmPERK 16* *ZmPERK 19* *ZmPERK 20* regardless of the tissues or stresses applied, they were always upregulated. We may conclude from these findings that *ZmPERK* gene expression is involved in drought stress and oil content accumulation in the embryo.

Table 3A
Predicted RNA editing sites in chloroplast genome.

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
ZmPERK1	14	5	TCC (S) → CTC (F)	ZmPERK7	134	45	CCC (P) → CTG (L)	ZmPERK13	319	107	CTC (L) → TTC (F)
	175	59	CCG (P) → TCG (S)		485	162	TCT (S) → TTT (F)		850	284	CCG (P) → TCG (S)
	203	68	CCG (P) → CTG (L)		514	172	CCG (P) → TCG (S)		853	285	CCC (P) → TCC (S)
	280	94	CCA (P) → TCA (S)		521	174	CCA (P) → CTA (L)		1187	396	ACC (T) → ATC (I)
	331	111	CCT (P) → TCT (S)		856	286	CAC (H) → TAC (Y)		1246	416	CTC (L) → TTC (F)
	338	113	CCG (P) → CTG (L)		949	317	CAC (H) → TAC (Y)		1379	480	CCA (P) → CTA (L)
	340	114	CCG (P) → TCG (S)		1034	345	GCA (A) → GTA (V)		1556	519	TCC (S) → TTC (F)
ZmPERK2	347	116	CCA (P) → CTA (L)	ZmPERK8	1225	409	CCC (P) → TTC (F)	ZmPERK14	1744	582	CAC (H) → TAC (Y)
	544	182	CAC (H) → TAC (Y)		1226	409	CCC (P) → TTC (F)		1877	626	ACC (T) → ATC (I)
	655	219	CAT (H) → TAT (Y)		236	79	CCC (P) → CTC (L)		379	127	GGG (R) → TGG (W)
	415	139	CCC (P) → TTC (F)		268	90	CCG (P) → TCG (S)		424	142	GTT (L) → TTT (F)
	416	139	CCC (P) → TTC (F)		313	105	CCT (P) → TCT (S)		442	148	CCA (P) → TCA (S)
	461	154	TCA (S) → TTA (L)		812	271	TCC (S) → TTG (L)		881	294	GCC (A) → GTC (V)
	577	193	CCC (P) → TCC (S)		1229	410	GCT (A) → GTT (V)		896	299	CCG (P) → CTG (L)
ZmPERK3	742	248	CCG (P) → TCG (S)	ZmPERK9	1277	426	GCT (A) → GTT (V)	ZmPERK15	326	109	CCG (P) → CTG (L)
	778	260	CAC (H) → TAC (Y)		1307	436	ACT (T) → ATT (I)		421	141	CCG (P) → TCG (S)
	841	281	CAC (H) → TAC (Y)		604	202	CCG (P) → TCG (S)		676	226	CAC (H) → TAC (Y)
	979	327	CAT (H) → TAT (Y)		865	289	CAC (H) → TAC (Y)		1034	345	GCG (A) → GTC (V)
	1093	365	CAC (H) → TAC (Y)		958	320	CAC (H) → TAC (Y)		1187	396	GCC (A) → GTC (V)
	863	288	ACC (T) → ATC (I)		997	333	CCG (P) → TCG (S)		1241	414	CCG (P) → CTG (L)
	1036	346	CCT (P) → TCT (S)		1063	355	CCC (P) → TCC (S)		1244	415	ACG (T) → ATG (M)
ZmPERK4	1103	368	CCT (P) → CTT (L)	ZmPERK10	1457	486	TCC (S) → TTC (F)	ZmPERK16	1306	436	CCC (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1187	396	TGC (S) → TTG (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1195	399	CAT (H) → TAT (Y)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1214	405	CCG (P) → CTG (L)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
					842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK5	1036	346	CCT (P) → TCT (S)	ZmPERK11	1457	486	TCC (S) → TTC (F)	ZmPERK17	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK6	1036	346	CCT (P) → TCT (S)	ZmPERK12	1457	486	TCC (S) → TTC (F)	ZmPERK18	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK7	1036	346	CCT (P) → TCT (S)	ZmPERK13	1457	486	TCC (S) → TTC (F)	ZmPERK19	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK8	1036	346	CCT (P) → TCT (S)	ZmPERK14	1457	486	TCC (S) → TTC (F)	ZmPERK20	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK9	1036	346	CCT (P) → TCT (S)	ZmPERK15	1457	486	TCC (S) → TTC (F)	ZmPERK21	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK10	1036	346	CCT (P) → TCT (S)	ZmPERK16	1457	486	TCC (S) → TTC (F)	ZmPERK22	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK11	1036	346	CCT (P) → TCT (S)	ZmPERK17	1457	486	TCC (S) → TTC (F)	ZmPERK23	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK12	1036	346	CCT (P) → TCT (S)	ZmPERK18	1457	486	TCC (S) → TTC (F)	ZmPERK24	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK13	1036	346	CCT (P) → TCT (S)	ZmPERK19	1457	486	TCC (S) → TTC (F)	ZmPERK25	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK14	1036	346	CCT (P) → TCT (S)	ZmPERK20	1457	486	TCC (S) → TTC (F)	ZmPERK26	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)
	1187	396	TGC (S) → TTG (L)		556	186	CCT (P) → TCT (S)		529	177	CCG (P) → TCG (S)
	1195	399	CAT (H) → TAT (Y)		737	246	GCT (A) → GTT (V)		586	196	CCA (P) → TCA (S)
	1214	405	CCG (P) → CTG (L)		842	281	GGC (A) → GTC (V)		628	210	CCT (P) → TCT (S)
ZmPERK15	1036	346	CCT (P) → TCT (S)	ZmPERK21	1457	486	TCC (S) → TTC (F)	ZmPERK27	1306	436	CCC (P) → TTC (F)
	1103	368	CCT (P) → CTT (L)		19	7	CTT (L) → TTT (F)		1307	436	CCG (P) → TTC (F)
	1151	384	CCT (P) → CTT (L)		241	81	CAC (H) → TAC (Y)		1520	507	CCC (P) → CTC (L)

Table 3A (continued)

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
ZmPERK4	289	97	CCC (P) → TTC (S)	ZmPERK11	883	295	CCC (P) → TCG (S)	ZmPERK17	1021	341	CAC (H) → TAC (Y)
	482	161	CCG (P) → CTC (L)		1132	378	CTC (L) → TTC (F)		1327	443	CAT (H) → TAT (Y)
	491	164	CCG (P) → CTC (L)		1172	391	CCC (P) → CTG (L)		1379	460	GCA (A) → GTA (V)
	565	189	CAT (H) → TAT (Y)						1435	479	CTT (L) → TTT (F)
	577	193	CCC (P) → TTC (S)		178	60	CCC (P) → TCG (S)		1441	481	CTT (L) → TTT (F)
	619	207	CCG (P) → TCG (S)		181	61	CCA (P) → TCA (S)				
	626	209	ACA (T) → ATA (I)		211	71	CCC (P) → TCG (S)		106	36	CCG (P) → TCG (S)
	859	287	CAC (H) → TAC (Y)		259	87	CCT (P) → TTT (F)		121	41	CCT (P) → TTT (F)
	143	48	TCG (S) → TTG (L)		260	87	CCT (P) → TTT (F)		122	41	CCT (P) → TTT (F)
					302	101	TCT (S) → TTT (F)		167	56	TCA (S) → TTA (L)
ZmPERK5	238	80	CCG (P) → TCG (S)	ZmPERK12	352	118	CCG (P) → TCG (S)	ZmPERK18	329	110	CCG (P) → CTG (L)
	752	251	TCA (S) → TTA (L)		359	120	CCG (P) → CTG (L)		338	113	CCA (P) → CTA (L)
	1010	337	GCC (A) → GTC (V)		376	126	CCC (P) → TCC (S)		382	128	CCT (P) → TTT (F)
	1031	344	CCG (P) → CTC (L)		388	130	CCG (P) → TCG (S)		383	128	CCT (P) → TTT (F)
	1226	409	GCC (A) → GTC (V)						425	142	TCT (S) → TTT (F)
	1367	456	TCC (S) → TTC (F)		389	130	CCG (P) → CTG (L)				
	1394	465	CCG (A) → GTC (V)		394	132	CCG (P) → TCG (S)		863	288	ACT (T) → ATT (I)
					430	144	CCG (P) → TCC (S)		1036	346	CCT (P) → TCT (S)
	275	92	ACT (T) → ATT (I)		535	179	CCA (P) → TCA (S)		1150	384	CCT (P) → TCT (S)
	400	134	CCT (P) → TTT (F)		542	181	CCG (P) → CTG (L)		1190	397	CCT (P) → CTT (L)
ZmPERK6	401	134	CCT (P) → TTT (F)	ZmPERK22	571	191	CCG (P) → TCG (S)	ZmPERK23	1198	400	CAT (H) → TAT (Y)
	512	171	ACA (T) → ATA (I)		920	307	TCG (S) → TTG (L)		1274	425	ACC (T) → ATC (I)
	940	314	CCG (P) → TCC (S)		1105	369	CTC (L) → TTC (F)		1319	440	ACC (T) → ATC (I)
	1319	440	ACG (T) → ATG (M)		1369	457	CCT (P) → TCT (S)		1343	448	GCG (A) → GTG (V)
	1439	480	TCC (S) → TTC (F)		1490	497	ACC (T) → ATG (M)				

Table 3B
Predicted RNA editing sites in mitochondrial genome.

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
ZmPERK1	5	2	ACG (T) => ATG (M)	ZmPERK8	8	3	TCC (S) => TTC (F)	ZmPERK13	319	107	CTC (L) => TTC (F)	ZmPERK20	104	35	CCG (P) => CTG (L)
	38	13	TCG (S) => TTG (L)		31	11	CCG (P) => TCG (S)		850	284	CCG (P) => TCG (S)		121	41	CCC (P) => TCC (S)
	40	14	CCC (P) => TCC (S)		38	13	CCG (P) => CTG (L)		853	285	CCC (P) => TCC (S)		149	50	TCG (S) => TTG (L)
	82	28	CTC (L) => TTC (F)		80	27	GCC (A) => GTC (V)		1187	396	ACC (T) => ATC (I)		182	61	GCA (A) => GTA (V)
	185	62	CCT (P) => CTT (L)		103	35	CCC (P) => TCC (S)		1246	416	CTC (L) => TTC (F)		239	80	CCA (P) => CTA (L)
	215	72	CCC (P) => CTC (L)		107	36	GCA (A) => GTA (V)		1379	460	CCA (P) => CTA (L)		272	91	TCT (S) => TTT (F)
	284	95	CCC (P) => CTC (L)		139	47	CAC (H) => TAC (Y)		1556	519	TCC (S) => TTC (F)		277	93	CCA (P) => TCA (S)
	305	102	ACC (T) => ATC (I)		215	72	CCA (P) => CTA (L)		1744	582	CAC (H) => TAC (Y)		284	95	GCC (A) => GTC (V)
	314	105	GCA (A) => GTA (V)		250	84	CCG (P) => TCG (S)		1877	626	ACC (T) => ATC (I)		305	102	CCA (P) => CTA (L)
	328	110	CCT (P) => TCT (S)		278	93	TCG (S) => TTG (L)		1925	642	GCC (A) => GTC (V)		320	107	GCG (A) => GTG (V)
	332	111	CCT (P) => CTT (L)		302	101	GCC (A) => GTC (V)		2045	682	CCC (P) => CTC (L)		329	110	GCA (A) => GTA (V)
	362	121	GCG (A) => GTG (V)		308	103	GCA (A) => GTA (V)		2054	685	GCT (A) => GTT (V)				
					329	110	CCT (P) => CTT (L)					ZmPERK21	535	179	CCT (P) => TCT (S)
ZmPERK2	389	130	CCT (P) => CTT (L)		341	114	GCA (A) => GTA (V)	ZmPERK14	92	31	ACG (T) => ATG (M)		545	182	CCG (P) => CTG (L)
	391	131	CCA (P) => TCA (S)		428	143	CCG (P) => CTG (L)		170	57	TCA (S) => TTA (L)		550	184	CCA (P) => TCA (S)
	406	136	CCG (P) => TCG (S)						386	129	CCT (P) => CTT (L)		566	189	TCT (S) => TTT (F)
	440	147	CCA (P) => CTA (L)	ZmPERK9	164	55	CCG (P) => CTG (L)		394	132	CAT (H) => TAT (Y)		641	214	CCT (P) => CTT (L)
	469	157	CCG (P) => TCG (S)		166	56	CCA (P) => TCA (S)		752	251	GCA (A) => GTA (V)		749	250	GCG (A) => GTG (V)
	512	171	TCA (S) => TTA (L)		230	77	TCG (S) => TTG (L)		923	308	GCA (A) => GTA (V)		815	272	GCG (A) => GTG (V)
	566	189	ACG (T) => ATG (M)		254	85	GCT (A) => GTT (V)		950	317	GCC (A) => GTC (V)		829	277	CCG (P) => TCG (S)
	572	191	TCA (S) => TTA (L)		389	130	GCA (A) => GTA (V)						836	279	CCT (P) => CTT (L)
	578	193	CCC (P) => CTC (L)		457	153	CCG (P) => TCG (S)	ZmPERK15	32	11	CCG (P) => CTG (L)		857	286	CCG (P) => CTG (L)
	808	270	CCG (P) => TCG (S)		671	224	TCG (S) => TTG (L)		106	36	CCC (P) => TCC (S)		935	312	CCG (P) => CTG (L)
	877	293	CCT (P) => TCT (S)		803	268	GCC (A) => GTC (V)		230	77	GCG (A) => GTG (V)				

Table 3B (continued)

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
	881	294	TCG (S) => TTG (L)		922	308	CTT (L) => TTT (F)		233	78	GCG (A) => GTG (V)	ZmPERK22	34	12	CAC (H) => TAC (Y)
					958	320	CAC (H) => TAC (Y)		281	94	GCC (A) => GTC (V)		76	26	CCA (P) => TCA (S)
ZmPERK3	223	75	CTT (L) => TTT (F)		1358	453	CCC (P) => CTC (L)		326	109	CCG (P) => CTG (L)		205	69	CCC (P) => TCC (S)
	317	106	GCT (A) => GTT (V)		1741	581	CAA (Q) => TAA (X)		404	135	GCG (A) => GTG (V)		667	223	CCT (P) => TCT (S)
	931	311	CCT (P) => TCT (S)						428	143	CCG (P) => CTG (L)		836	279	CCA (P) => CTA (L)
	1198	400	CAC (H) => TAC (Y)	ZmPERK10	5	2	ACG (T) => ATG (M)		430	144	CCG (P) => TCG (S)		890	297	CCA (P) => CTA (L)
	1277	426	GCC (A) => GTC (V)		8	3	CCG (P) => CTG (L)		439	147	CCG (P) => TCG (S)		1000	334	CAT (H) => TAT (Y)
	1322	441	ACC (T) => ATC (I)		10	4	CCG (P) => TCG (S)		488	163	GCG (A) => GTG (V)		1033	345	CCC (P) => TCC (S)
	1373	458	CCA (P) => CTA (L)		31	11	CCG (P) => TCG (S)		571	191	CCC (P) => TCC (S)		1040	347	GCT (A) => GTT (V)
	1382	461	ACC (T) => ATC (I)		98	33	GCG (A) => GTG (V)		614	205	GCC (A) => GTC (V)		1181	394	TCG (S) => TTG (L)
					476	159	GCG (A) => GTG (V)		659	220	ACC (T) => ATC (I)				
ZmPERK4	8	3	TCT (S) => TTT (F)		608	203	GCT (A) => GTT (V)		748	250	CCC (P) => TCC (S)	ZmPERK23	125	42	CCG (P) => CTG (L)
	31	11	CCA (P) => TCA (S)		614	205	ACA (T) => ATA (I)						155	52	CCG (P) => CTG (L)
	38	13	CCG (P) => CTG (L)		884	295	CCG (P) => CTG (L)	ZmPERK16	235	79	CCC (P) => TTC (F)		191	64	TCG (S) => TTG (L)
	41	14	TCT (S) => TTT (F)						236	79	CCC (P) => TTC (F)		218	73	GCA (A) => GTA (V)
	74	25	TCT (S) => TTT (F)	ZmPERK11	34	12	CCT (P) => TCT (S)		302	101	CCA (P) => CTA (L)		395	132	GCC (A) => GTC (V)
	89	30	ACT (T) => ATT (I)		77	26	ACG (T) => ATG (M)		331	111	CCG (P) => TCG (S)		944	315	CCA (P) => CTA (L)
	107	36	GCG (A) => GTG (V)		176	59	ACT (T) => ATT (I)		425	142	ACG (T) => ATG (M)		1165	389	CCT (P) => TTT (F)
	137	46	CCC (P) => CTC (L)		194	65	CCG (P) => CTG (L)		571	191	CCG (P) => TCG (S)		1166	389	CCT (P) => TTT (F)
	164	55	GCG (A) => GTG (V)		224	75	CCC (P) => CTC (L)		626	209	CCT (P) => CTT (L)		1405	469	CCG (P) => TCG (S)
	182	61	GCT (A) => GTT (V)		232	78	CCA (P) => TCA (S)		632	211	GCT (A) => GTT (V)		1739	580	GCT (A) => GTT (V)
	197	66	CCA (P) => CTA (L)		239	80	GCT (A) => GTT (V)		815	272	TCG (S) => TTG (L)				
	272	91	CCC (P) => CTC (L)		254	85	CCT (P) => CTT (L)		1021	341	CAC (H) => TAC (Y)				
					260	87	CCT (P) => CTT (L)		1124	375	GCG (A) => GTG (V)				

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Table 3B (continued)

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
ZmPERK5	208	70	CCC (P) => TCC (S)		289	97	CCT (P) => TTT (F)		1228	410	CAT (H) => TAT (Y)				
	236	79	CCG (P) => CTG (L)		290	97	CCT (P) => TTT (F)		1277	426	GCT (A) => GTT (V)				
	257	86	ACC (T) => ATC (I)		340	114	CCA (P) => TCA (S)								
	577	193	CCG (P) => TCG (S)		347	116	GCA (A) => GTA (V)	ZmPERK17	34	12	CAC (H) => TAC (Y)				
	686	229	GCG (A) => GTG (V)		371	124	CCG (P) => CTG (L)		76	26	CCA (P) => TCA (S)				
	943	315	CCA (P) => TCA (S)		383	128	GCC (A) => GTC (V)		205	69	CCC (P) => TCC (S)				
	1010	337	GCC (A) => GTC (V)		395	132	CCT (P) => CTT (L)		667	223	CCT (P) => TCT (S)				
	1094	365	GCG (A) => GTG (V)						836	279	CCA (P) => CTA (L)				
			ZmPERK12	266	89		TCG (S) => TTG (L)		890	297	CCA (P) => CTA (L)				
ZmPERK6	49	17	CCA (P) => TCA (S)		268	90	CCG (P) => TCG (S)		1000	334	CAT (H) => TAT (Y)				
	64	22	CCT (P) => TTT (F)		296	99	CCT (P) => CTT (L)		1033	345	CCC (P) => TCC (S)				
	65	22	CCT (P) => TTT (F)		299	100	CCT (P) => CTT (L)		1040	347	GCT (A) => GTT (V)				
	92	31	ACA (T) => ATA (I)		304	102	CCG (P) => TCG (S)		1181	394	TCG (S) => TTG (L)				
	775	259	CTT (L) => TTT (F)		332	111	CCG (P) => CTG (L)								
	1316	439	TCA (S) => TTA (L)		356	119	GCG (A) => GTG (V)	ZmPERK18	55	19	CCC (P) => TTC (F)				
	1412	471	ACA (T) => ATA (I)		359	120	GCG (A) => GTG (V)		56	19	CCC (P) => TTC (F)				
	1420	474	CCG (P) => TCG (S)		386	129	GCA (A) => GTA (V)		62	21	GCC (A) => GTC (V)				
	1441	481	CCG (P) => TCG (S)		395	132	CCG (P) => CTG (L)		83	28	GCC (A) => GTC (V)				
					440	147	GCC (A) => GTC (V)		223	75	CTC (L) => TTC (F)				
ZmPERK7	65	22	CCG (P) => CTG (L)		449	150	ACG (T) => ATG (M)		317	106	GCT (A) => GTT (V)				
	109	37	CTT (L) => TTT (F)		461	154	TCA (S) => TTA (L)		931	311	CCT (P) => TCT (S)				
	137	46	GCC (A) => GTC (V)		476	159	GCG (A) => GTG (V)		1127	376	TCA (S) => TTA (L)				
	143	48	GCG (A) => GTG (V)		521	174	ACC (T) => ATC (I)		1226	409	GCT (A) => GTT (V)				
	257	86	GCT (A) => GTT (V)		524	175	GCC (A) => GTC (V)		1370	457	CCA (P) => CTA (L)				
	260	87	CCA (P) => CTA (L)		535	179	CCA (P) => TCA (S)		1379	460	ACC (T) => ATC (I)				

Table 3B (continued)

Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation	Genes	Nucleotide Position	Amino acid Position	Amino acid Conservation
	275	92	GCG (A) => GTG (V)		616	206	CCA (P) => TCA (S)								
	290	97	TCA (S) => TTA (L)		620	207	CCT (P) => CTT (L)	ZmPERK19 4		2	CCC (P) => TTC (F)				
	311	104	GCG (A) => GTG (V)		644	215	CCG (P) => CTG (L)		5	2	CCC (P) => TTC (F)				
	320	107	GCT (A) => GTT (V)		674	225	TCG (S) => TTG (L)		26	9	CCG (P) => CTG (L)				
	323	108	GCT (A) => GTT (V)						38	13	TCG (S) => TTG (L)				
	524	175	ACA (T) => ATA (I)						43	15	CCG (P) => TCG (S)				
	614	205	CCC (P) => CTC (L)						59	20	TCT (S) => TTT (F)				
	770	257	GCG (A) => GTG (V)						70	24	CTT (L) => TTT (F)				
									98	33	GCG (A) => GTG (V)				
									194	65	CCG (P) => CTG (L)				
									200	67	CCG (P) => CTG (L)				
									299	100	GCC (A) => GTC (V)				

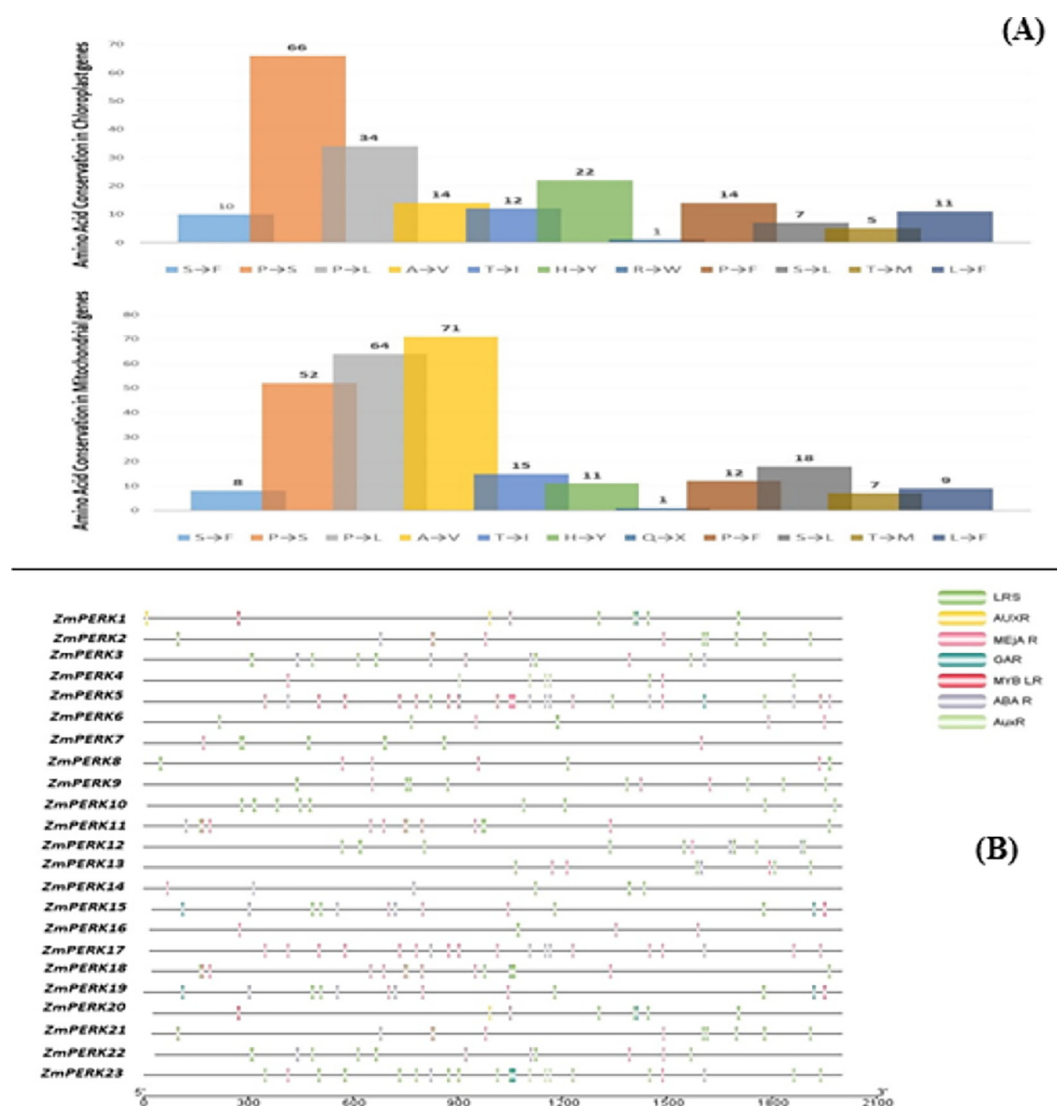


Fig. 6. (A) RNA editing of the *PERK* genes results in amino acid conservation. (B) Identified *cis*-acting elements in *ZmPERK* gene family promoters.

4. Discussion

Many ancient land plants evolved over the time also possess *PERK* genes (Nakhamchik et al., 2004, Qanmber et al., 2019, Chen et al., 2020) which depict that these gene families are present from centuries in the plants. Now a day's modern techniques of DNA sequences have revolutionized the field of DNA sequencing especially the advent of next-generation sequencing technologies, have shorten the time of sequencing with more accuracy in the results. The availability of maize genome assemblies has open the new avenues for studying various functions of genes at genome-wide level. There has been no systematic study of maize to date. However, in our study we discovered 23 *ZmPERK* genes in the maize genome. In phylogenetic analysis, we divided *ZmPERK* genes into four groups. The findings demonstrated that *PERK* genes were initially originated in ancient land plants and their orthologous genes may be found throughout the plant kingdom. Plant *PERK* genes from dicot, monocot, lycopphytes, and chlorophytes were assigned to each of the four clades randomly. Our study showed that *ZmPERK* genes remained evolutionarily conserved, as these were found in each of the species which was used in this study. Furthermore, the expansion of these gene into higher plants

was occurred with the passage of time. According to sequence logos for *PERK* genes, the protein sequence residues were highly conserved, and no compositional bias was seen across the studied species. To study the evolutionary history of multiple gene families (Ohta, 2010). It is essential to know the structure of the genes. The length of nucleotide and amino acid sequences varied considerably, indicating that the *ZmPERK* genes are diverse. It is quite worth to study the exon–intron structure as insertion/deletion events play important role in determine the structure of exon–intron. The introns gain or lose have been witnessed throughout eukaryotic diversification. The exon–intron pattern of duplicated genes is similar, whereas more diversification observed in the intron length suggesting that the intron length may be significant in *ZmPERK* functional diversification. In *ZmPERK* proteins, different combinations of conserved motifs were identified. Protein motif analysis revealed that protein from same species were tend to fall in the same cluster together. Arrangements of motif were comparable among members of the same subfamily.

The study of gene duplication events is critical as these play crucial role in the genome expansions and alignments (Tamura et al., 2011). The gene duplication events have been witnessed in various transcription factor families of plants (Liu et al., 2011;

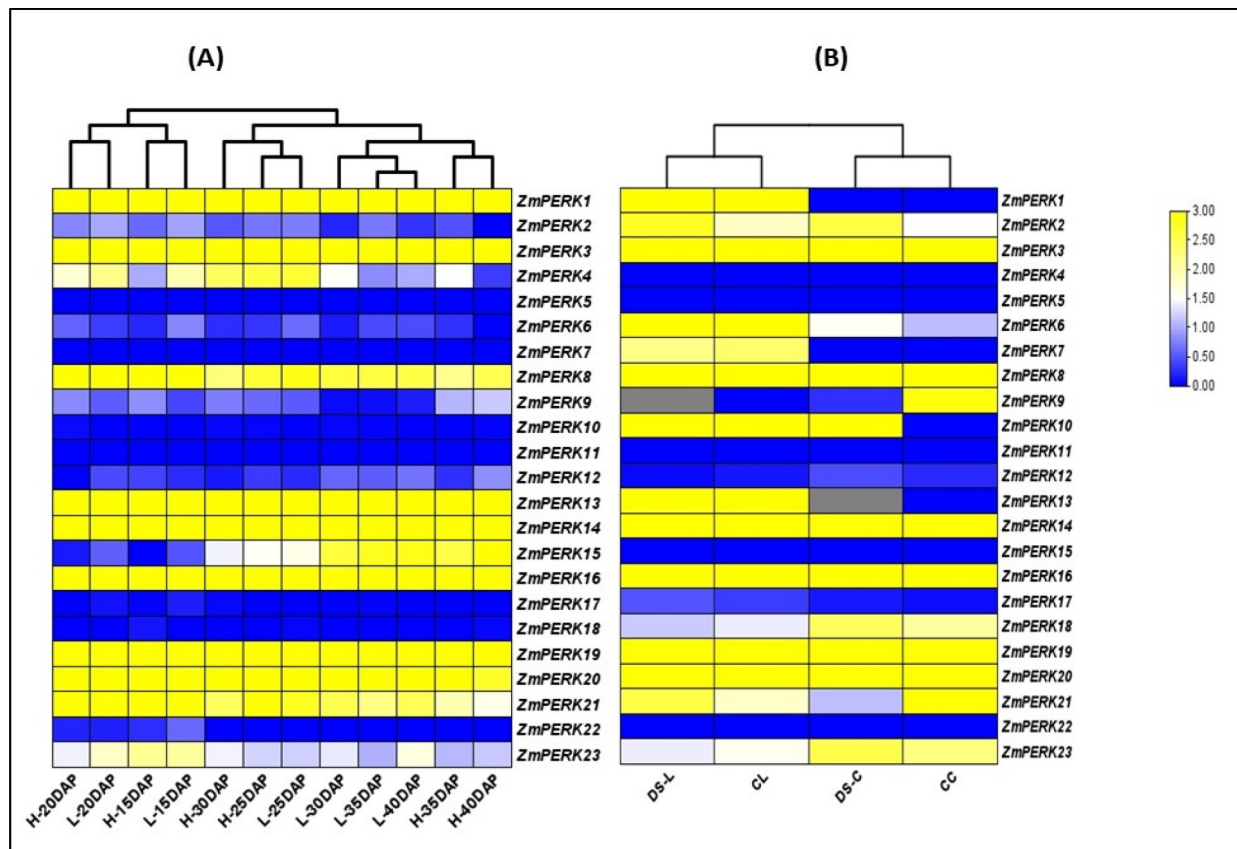


Fig. 7. A: *ZmPERK* gene expression patterns at different developmental stages of seed embryos from 15 days after pollination (DAP) to 40 days after pollination (DAP). H represent expression in high oil content variety while L represent low oil content variety B: Expression pattern of *ZmPERK* gene under drought stress DS-L (Drought stress leave sample) DS-C (Drought stress cob sample) CL (Controlled leave sample) CC (Control Cob sample).

Shan et al., 2013). To differentiate whether the gene duplication was the result of tandem or segmental if the duplications are the result of the presence two or more genes the same chromosome, it will be considered as tandem duplication whereas segmental or WGD duplications are when two or more genes are duplicated on different chromosomes. The intron expansion is mainly the result of the tandem duplications and thus give rise to the formation of the new genes (Yang et al., 2008), but we only find evidence of one tandem and eight segmental duplications in this study. In Plants the environmental and selection factors have expanded multiple gene families more than other eukaryotes organisms. The *Ka/Ks* ratios demonstrated that maize *PERK* genes have been subjected to extensive selection, with relatively minimal functional variations due to whole genome and segmental duplication.

ZmPERK genes possess *cis*-elements associated with stress responses in their promoter regions. *ZmPERK* contains *cis*-elements such as MeJA responsive, MYB-binding sites associated with light responsiveness elements, ABA responsive, defense, and stress responsive, low temperature and gibberellin acid (GA) responsive elements. The presence of these *cis* elements with specified characteristics demonstrated the putative role in plant growth, development, as well as in biotic and abiotic stress response. Synteny is a framework for assessing homologous gene and gene order conservation across genomes of different species. The collinearity between maize and sorghum was shown to be more significant than the collinearity between maize and rice.

Plant growth and development are assisted by RNA editing, which is an effective strategy for regulating gene expression at the post-transcriptional level in higher plant organelle genomes. The discovery and identification of RNA editing sites is critical for

a better knowledge of their biological activities and establishing the framework for future research and comprehension of their molecular processes. In this work, the RNA editing sites of chloroplast and mitochondrial genes in maize were predicted. Table 3A lists 196 RNA editing sites predicted in chloroplast genes and 268 in mitochondrial genes (Table 3B). In the chloroplast and mitochondrial genomes, these sites were detected on 23 genes, with an average of 8.5 and 11.26 editing sites per gene, respectively. The transition and conservation of cytosine (C) to uracil (U) was observed in all of the editing sites. Changes in the first and second codon nucleotides were mostly involved for these transitions. The current study laid the groundwork for future research into the biological functions of chloroplast and mitochondrial RNA editing in maize. The expression patterns of genes are closely related to their biological functions. According to expression analyses *ZmPERK* genes were shown to be important in drought tolerance and oil content accumulation in embryos.

5. Conclusion

The current study found 23 non-redundant *ZmPERK* encoding genes in maize. The *PERK* gene family is conserved among the analyzed plant species, according to their classification, characterization in terms of gene structure, motif, conserved domains, and comparative phylogenetic analyses. Furthermore, gene duplication analysis and syntenic relationship studies reveal that the maize paralogous genes proliferate through segmental duplications, whereas codons went under purifying selection, resulting in a significant expansion of the *ZmPERK* gene family. The existence of

putative *cis*-elements in the *ZmPERK* gene promoter regions suggests that they have a functional role in growth, development, and stress resilience. Most of the genes were found to be up regulated in response to stress and oil content, accumulation showing that they may play a role in stress modulation and development process in maize. Overall, these findings will assist in the functional characterization of maize PERK genes. The candidate *ZmPERK* genes can be employed in a breeding program.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jksus.2022.102293>.

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